CSA0711

(6)

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0711

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I CH. Abhilash reddy (192525462) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Ch. Abhilash reddy

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5 10marks
Problem Understanding	20%	1.6	1.6	1.6	1.6	1.4
Method Selection	20%	1.6	1.6	1.6	1.6	1.4
Solution Process	25%	2.0	2.0	2.0	2.0	1.75
Accuracy of Calculation	20%	1.6	1.6	1.6	1.6	1.4
Interpretation of Results	15%	1.2	1.2	1.2	1.2	1.05

Signature of the Faculty

Total Marks /40 39

39/100

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CO2 - A11

A university Network is assigned the IP address block 192.168.20.0/24 for its internal communication. The Network as the adminstartor has decided to divide this Network into subsets for three different labs using fixed subnet masks.

Subnetting 192.168.20.0/24 for three labs

Introduction:-
Subnetting is the process of dividing one large IP network into smaller logical Networks called subnets. It improves the Network performance, security, address management, and reduces unnecessary broadcast traffic.

The university is give the Network:
192.168.20.0/24

A/24 Network must be divided among three labs using different Subnet sizes

Lab A → /26
Lab B → /27
Lab C → /28

The subnets are allocated continuously without overlapping

Important Formulas

Total addresses

$$2^{32-\text{Prefix length}}$$

Usable host addresses

$$2^{32-\text{Prefix length}} - 2$$

Two addresses are normally reserved:

* First address → Network address
* Last address → Broadcast address

Lab A - /26 subnet

A /26 prefix uses 26 bits for the network and 6-bit for host

Subnet mask

255.255.255.192

Total addresses

$2^{32-26} = 2^6 = 64$

Usable hosts

$64 - 2 = 62$

Block size

$256 - 192 = 64$

therefore, the first subnet range is:-

192.168.20.0 to 192.168.20.63.

Network address :- 192.168.20.0
First valid host:- 192.168.20.1
Last valid host:- 192.168.20.
Broadcast address :- 192.168.20.63
usable hosts :- 62

Lab B - /27 - subnet

The Next available address after Lab A is:

192.168.20.64

Subnet masks:- 255.255.255.224.

Total addresses :-

$2^{32-27} = 2^5 = 32$

usable hosts :-

$32 - 2 = 30$

Block size :-

$256 - 224 = 32.$

herefore, the subnet range is
192.168.20.64 to 192.168.20.95

etwork address:- 192.168.20.64

First valid host:- 192.168.20.65

last valid host:- 192.168.20.94

Broadcast address :- 192.168.20.95

usable hosts :- 30

lab - C - /28 - subnet

The next available address after Lab B is:

192.168.20.96

Subnet masks

255.255.255.240

lab	prefix	subnet masks	Network address	Valid Host Range	Broadcast	
lab A	/26	255.255.255.192	192.168.20.0	192.168.20.1 192.168.20.62	192.168.20.63	62
lab B	/27	255.255.255.224	192.168.20.64	192.168.20.65		30
		255.255.255.240	192.168.20.96	192.168.20.97		14

Q2 Subnet Analysis of Two System

Given Information

System 1:-

IP address :- 192.168.1.130

subnet masks:- 255.255.255.192

System 2:-

IP address :- 192.168.1.190

subnet masks:- 255.255.255.192

The subnet mask 255.255.255.192 is Equivalent to /26

Determine the Block size.

The fourth octect of the subnet mask is 192

256 - 192 = 64

192.168.1.0 - 192.168.1.63

192.168.1.64 - 192.168.1.127

192.168.1.128 - 192.168.1.191

192.168.1.192 - 192.168.1.255

Determine the subnet for system 1.

System 1 has the IP

192.168.1.130

The address fall between:-

192.168.1.128 and 192.168.1.191

Therefore, system 1 belongs to:

192.168.1.128/26

System 2 subnet details

Network address:- 192.168.1.128

First valid host :- 192.168.1.129.

Last valid host :- 192.168.1.190.

Broadcast address :- 192.168.1.191.

Host calculation:-

A/26 Network leaves:

$32 - 26 = 6$

Total address

$= 2^6 = 64$

$64 - 2 = 62$

Final comparsion:-

Detail	System 1	System 2
IP address		192.168.1.190
Prefix		/26
Network address		192.168.1.128
Host range		192.168.1.129 – 192.168.1.1
Broardcast address		192.168.1.191
Valid host?	Yes	Yes

3 Department subnetting of 172.16.0.0/16

Introduction

The organization owns the Network:

172.16.0.0/16

A/16 Network contains:

$2^{32-16} = 65,536$

HR → /25

Sales → /26

IT → /27

Admin → /28

HR Department - /25

Subnet mask

255.255.255.128

$2^{7} = 128$

$128 - 2 = 126$

This first available subnet is

192. 16.0./25

Sales - Department - /26

The next available address is:

172.16.0.128

A/26 Subnet has 6 hosts bits

255.255.255.192

$2^6 = 64$

usable hosts

$64 - 2 = 62$

The Subnet is:-

172.16.0.128/26

IT Department /-27

172.16.0.192

A/27 Subsent has 5 hosts bits

255.255.255.224

$2^5 = 32$

$32 - 2 = 30$

The subnet is:

172.16.0.192/27

Network address :- 192.16.0.192

First hosts :- 192.16.0.193

last hosts :- 172.16.0.222

Broadcast address :- 172.16.0.223

Usable hosts :- 30

Admin Department - 128

he next available address is:
172.26.0.224

A128 subnet has 4 hosts bits

255.255.255.240.

Department	prefix	Network address	valid Host Range
HR	/25	172.16.0.1 - 172.16.0.126	192.16.0.127
Sales	/26	172.16.0.129 - 172.16.0.190	172.16.0.191
IT	/27	172.16.0.193 - 172.16.0.222	172.16.0.223
Admin	/28	172.16.0.225 - 172.16.0.238	172.16.0.239

Q. Subnetting 172.16.10.0/24 using /27

Given

Original Network: 172.16.10.0/24

New prefix : /27

The tasks is to find:-

Number of students

unabale hosts per subnet

Subnet containg 172.16.10.78

Network and bradcast address

Whether 172.16.10.95 belongs to the same subnet.

Number of subnets

Number of borrowed bits:-

$27 - 24 = 3$

Number of subnets:-

$2^3 = 8$

Total number of subnets = 8

Hosts per subnet

$32 - 27 = 5$

$2^5 = 32$

$32 - 2 = 30$

Unable hosts per subnet = 30

Determine the block size

255.255.255.224

Block size:-

$256 - 224 = 32$

Check 172.16.10.95

172.16.10.64 to 172.16.10.95

IPV6 Address compression, Expansion and /64 prefix

Given IPV6 Address:-

2001: odb8: 0000: 0000: 0000: ff00: 0042: 8329

emove Leading zeros:-

2001: odb8: 0000: 0000: 0000: ff00: 0042: 8329

After removing leading zeros

Examples:-

odb8 becomes db8

0000 becomes 0

0042 becomes 42

Apply zero compression:-

the address contain three consecutive zero groups

0: 0: 0

therefore, the compressed address is

2001: db8: ff00:: 42: 8329

To Expand it, count the visible groups

2001
db8
ff00
42
8329

These are five visible groups. since a full IPV6 address require eight groups, three zero groups must be inserted

Therefore, the full expanded form is

2001: 0db8: 0000: 0000: 0000: ff00: 0042: 8329

Determine the 164 Network prefix

A/64 prefix means:

$4 \times 16 = 64$

The first four groups are:

2001:0db8:0000:0000

Thus, the network prefix is

2001:0db8:0000:0000::/64

compressed form

2001:db8::/64

Host Portion

The last groups are

0000:ff00:0042:8329

with a 164 prefix:

$128 - 64 = 64$

2^{64}

$= 18,446,744,073,709,551,616$

Final answer

Answer

original address

2001:0bd8:0000:0000:0000:0000:ff00:0042:8329

compressed form

2001::db8:ff00:42:8329

Network bits

64

Host bits

64